8.3.2 Identifying the components of the platform

In our architecture, the platform interface and the kernel are largely composed of Databricks and related services. The team of one could start to provide templates or some kind of service for provisioning of the Kinesis and Fargate components, but that would probably clash with the idea of having a very lean platform team.

What the platform team will have to do is to provide guidelines, a one-pager of how to use Databricks inside the data mesh. These guidelines will indicate how to work with access rights, where to store all data products, and the likes.

8.3.3 Identifying the components of the data product

All data products are built on top of Databricks. Let’s describe them in detail.

The data product owned by the Black team is a bit more involved. The upstream data is sourced from an operational RDS instance and some other systems. The team uses a custom Fargate task to load the data periodically into an S3 bucket. From there, a Databricks Spark batch job picks up the data and shoves it into Delta Lake. Then a Databricks notebook is used to transform the data into a data product and put it again into Delta Lake, from where it is exposed to others to use. The Delta Lake dataset in this case forms the data output port.

The data product owned by the White team uses two upstream datasets, as you can see in figure 8.10. A user periodically inputs data into an S3 bucket, which is picked up by a batch job and delivered into Delta Lake for temporary storage. This forms a push interface for the data product. On the other side, the team has provided a custom Kinesis Data Stream. Operational components push data in real time into the stream, which then is picked up by a Databricks streaming job. Again, a notebook is used for transformation and to produce the data output.